Elementary Analysis

Chapter 2, Section 14

Series

Exercises 14.1–14.14, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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14 Series

For a sequence (a_n), its Nth partial sum is $S_N = \sum_{n=1}^{N} a_n$. The series $\sum_{n=1}^{\infty} a_n$ converges precisely when (S_N) has a finite real limit. A limit of $+\infty$ or $-\infty$ is divergence, even though it specifies a direction of divergence.

The necessary condition $a_n \to 0$ is not sufficient for convergence. The Comparison, Ratio, and Root Tests supply sufficient conditions in the situations specified below. When terms are nonnegative, convergence and absolute convergence are the same assertion. Unless a lower index is explicitly given, the series in the catalogs begin at $n = 1$; changing finitely many initial terms does not affect convergence, as proved in Exercise 14.9. The logarithm is the natural logarithm.

14.1 Exercise 14.1: Six initial convergence tests

What we must show. Decide convergence for each of the six given series, with a reason that checks the hypotheses of the chosen test.

(a) The series with term $n^4/2^n$.

Let $a_n = n^4/2^n > 0$. Since no term is zero, the successive ratio is well defined. Canceling powers gives

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^4}{2^{n+1}} \frac{2^n}{n^4} = \frac{1}{2} \left(1 + \frac{1}{n}\right)^4 \longrightarrow \frac{1}{2}.$$

The last limit follows from $1/n \to 0$ and the finite product limit law. Its value is strictly below 1, so the Ratio Test proves absolute convergence. Since the terms are positive, this proves convergence of the requested series.

(b) The series with term $2^n/n!$.

Put $a_n = 2^n/n! > 0$. The identity $(n+1)! = (n+1)n!$ yields

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \frac{n!}{2^n} = \frac{2}{n+1} \longrightarrow 0.$$

Here $0 < 1$, so the Ratio Test proves absolute convergence.

(c) The series with term $n^2/3^n$.

With $a_n = n^2/3^n > 0$, the same cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{1}{3} \left(1 + \frac{1}{n}\right)^2 \longrightarrow \frac{1}{3} < 1.$$

The Ratio Test again gives absolute convergence. The power of n in the numerator does not prevent this conclusion because the computed ratio limit is strictly less than 1.

(d) The series with term $n!/(n^4+3)$.

Let $a_n = n!/(n^4+3) > 0$. Its ratio is

$$\frac{a_{n+1}}{a_n} = (n+1) \frac{n^4+3}{(n+1)^4+3} = (n+1) \frac{1 + 3/n^4}{(1+1/n)^4 + 3/n^4}.$$

The second factor tends to 1, since its numerator and denominator both tend to 1 and the denominator is never zero. In particular, it exceeds $1/2$ eventually. Hence the displayed ratio is eventually at least $(n+1)/2$, and therefore tends to $+\infty$. The divergent part of the Ratio Test applies, since the ratio limit inferior is greater than 1.

We can also see why the term condition fails. There is N after which every ratio is at least 2. Induction then gives $a_{N+k} \geq 2^k a_N$, with $a_N > 0$, so the terms tend to $+\infty$ rather than to zero. The positive partial sums are unbounded, and the series diverges to $+\infty$.

(e) The series with term $\cos^2 n/n^2$.

The trigonometric bound $-1 \leq \cos n \leq 1$ gives $0 \leq \cos^2 n \leq 1$. Since $n^2 > 0$, division preserves order:

$$0 \leq \frac{\cos^2 n}{n^2} \leq \frac{1}{n^2}.$$

The comparison series $\sum 1/n^2$ converges by the p-series criterion with $p = 2 > 1$. The Comparison Test therefore proves convergence of the given nonnegative series.

(f) The series with term $1/\log n$, beginning at $n = 2$.

For $n \geq 2$, the logarithm is positive. The elementary logarithm bound is

$$0 < \log n = \int_1^n \frac{dt}{t} \leq \int_1^n 1\,dt = n - 1 < n.$$

The integral comparison here only justifies the inequality for one logarithm; no integral test for a series is being invoked. Taking reciprocals of positive quantities reverses the inequality, so $1/\log n \geq 1/n$. The harmonic series diverges to $+\infty$, and deleting its first term leaves divergent partial sums. Thus the Comparison Test proves divergence to $+\infty$.

Conclusion. Parts (a), (b), (c), and (e) converge absolutely. Parts (d) and (f) diverge to $+\infty$.

14.2 Exercise 14.2: Seven further convergence tests

What we must show. Classify the seven series, explaining the actual numerator and denominator in each term before applying a test.

(a) The series with term $(n-1)/n^2$.

All terms are nonnegative. For $n \geq 2$, the inequality $n - 1 \geq n/2$ follows by subtracting $n/2$ from both sides: $n/2 - 1 \geq 0$. Therefore

$$\frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n} \qquad (n \geq 2).$$

The series $\sum_{n=2}^{\infty} 1/(2n)$ is one half of a tail of the harmonic series, so it diverges to $+\infty$. The Comparison Test proves divergence to $+\infty$ of the requested series. The first term is zero and has no effect on this lower bound for the later partial sums.

(b) The series with term $(-1)^n$.

Every term has absolute value 1, so the terms do not tend to zero. Corollary 14.5 therefore rules out convergence. Directly, the partial sums are $-1, 0, -1, 0, \ldots$: an odd number of terms gives -1, and

each pair $(-1)+1$ sums to zero. Thus the partial sums have two different subsequential limits and do not tend to either infinity.

(c) The series with term $3n/n^3$.

Since $n \neq 0$, cancellation gives $3n/n^3 = 3/n^2$. The p-series $\sum 1/n^2$ converges, and multiplication of all partial sums by the constant 3 preserves convergence by the constant multiple limit law. Hence this positive series converges absolutely.

(d) The series with term $n^3/3^n$.

For $a_n = n^3/3^n > 0$, compute

$$\frac{a_{n+1}}{a_n} = \frac{1}{3}\left(1+\frac{1}{n}\right)^3 \longrightarrow \frac{1}{3} < 1.$$

All ratios are defined; the Ratio Test gives absolute convergence.

(e) The series with term $n^2/n!$.

Put $a_n = n^2/n! > 0$. Cancel $n!$ and one factor of $n+1$:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(n+1)!}\frac{n!}{n^2} = \frac{n+1}{n^2} = \frac{1}{n} + \frac{1}{n^2} \longrightarrow 0 < 1.$$

The Ratio Test proves absolute convergence.

(f) The series with term $1/n^n$.

The terms are positive and their positive nth roots satisfy

$$\left(\frac{1}{n^n}\right)^{1/n} = \frac{1}{n} \longrightarrow 0.$$

Thus the limit superior of the roots is $0 < 1$. The convergent part of the Root Test proves absolute convergence.

(g) The series with term $n^n/2^n$.

Here the positive nth root is

$$\left(\frac{n^n}{2^n}\right)^{1/n} = \frac{n}{2} \longrightarrow +\infty.$$

The root limit superior is greater than 1, so the Root Test proves divergence. More explicitly, for $n \geq 4$ one has $n/2 \geq 2$, so $(n/2)^n \geq 2^n$. These positive terms tend to $+\infty$; hence the partial sums do too.

Conclusion. Parts (c), (d), (e), and (f) converge absolutely. Parts (a) and (g) diverge to $+\infty$. Part (b) diverges with bounded, oscillating partial sums.

14.3 Exercise 14.3: Factorials, geometric comparisons, and a periodic term

What we must show. Classify each of the six series and justify convergence or failure of the necessary term condition.

(a) The series with term $1/\sqrt{n!}$.

Set $a_n = 1/\sqrt{n!} > 0$. The factorial is inside the square root. Using $(n+1)! = (n+1)n!$ and positivity,

$$\frac{a_{n+1}}{a_n} = \frac{\sqrt{n!}}{\sqrt{(n+1)!}} = \frac{1}{\sqrt{n+1}} \longrightarrow 0.$$

The Ratio Test applies and gives absolute convergence.

(b) The series with term $(2+\cos n)/3^n$.

Since $-1 \le \cos n \le 1$, adding 2 gives $1 \le 2 + \cos n \le 3$. Consequently

$$0 \le \frac{2+\cos n}{3^n} \le \frac{3}{3^n}.$$

The comparison series is geometric, with positive constant factor 3 and ratio $1/3 < 1$. It converges, so the Comparison Test proves absolute convergence of the given nonnegative series.

(c) The series with term $1/(2^n + n)$.

We have $2^n + n \ge 2^n > 0$. Reciprocation yields

$$0 < \frac{1}{2^n + n} \le \frac{1}{2^n}.$$

The upper comparison is a geometric series of ratio $1/2$, so the requested series converges absolutely.

(d) The series with term $(1/2)^n(50 + 2/n)$.

For $n \ge 1$, the inequality $0 < 2/n \le 2$ implies $0 < 50 + 2/n \le 52$. Multiply by the positive number 2^{-n}:

$$0 < \left(\frac{1}{2}\right)^n \left(50 + \frac{2}{n}\right) \le \frac{52}{2^n}.$$

The series on the right is a constant multiple of a convergent geometric series. The Comparison Test therefore gives absolute convergence. In particular, the expression in parentheses is $50 + 2/n$, with n in the denominator of its second term.

(e) The series with term $\sin(n\pi/9)$.

Consider the strictly increasing indices $n_k = 18k + 3$, $k \in \mathbb{N}$. The corresponding terms are

$$\sin\left(\frac{(18k+3)\pi}{9}\right) = \sin(2k\pi + \pi/3) = \sin(\pi/3) = \frac{\sqrt{3}}{2}.$$

This nonzero constant subsequence prevents the whole term sequence from tending to zero: every subsequence of a sequence tending to zero would also tend to zero. Thus Corollary 14.5 proves that the series diverges. A term test does not require the terms to be nonnegative or the partial sums to tend to $+\infty$.

(f) The series with term $100^n/n!$.

For $a_n = 100^n/n! > 0$, factorial cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{100}{n+1} \longrightarrow 0 < 1.$$

The Ratio Test gives absolute convergence, despite the relatively large fixed base 100 in the numerator.

Conclusion. Every series in this exercise except (e) converges absolutely. Part (e) diverges because its terms do not tend to zero.

14.4 Exercise 14.4: Oscillating denominators, telescoping roots, and factorials

What we must show. Classify the three series, keeping track of the starting index and the placement of the powers.

(a) The series with term $1/[n+(-1)^n]^2$, for $n \geq 2$.

For these indices,

$$n + (-1)^n \geq n - 1 \geq \frac{n}{2} > 0.$$

The middle inequality is equivalent to $n \geq 2$. Because both sides are positive, squaring preserves order and taking reciprocals reverses it. We obtain

$$0 < \frac{1}{[n+(-1)^n]^2} \leq \frac{4}{n^2} \qquad (n \geq 2).$$

The comparison series converges by the p-series criterion with $p = 2$ and the constant multiple limit law. Hence the given series converges absolutely. Beginning at $n = 2$ also avoids the zero denominator that would occur at $n = 1$.

(b) The series with term $\sqrt{n+1} - \sqrt{n}$.

Each term is positive. Its Nth partial sum telescopes:

$$\begin{aligned}\sum_{n=1}^{N} (\sqrt{n+1} - \sqrt{n}) &= (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + \cdots + (\sqrt{N+1} - \sqrt{N}) \\ &= \sqrt{N+1} - 1.\end{aligned}$$

All intermediate square roots cancel exactly in this finite sum. For a given $M > 0$, any $N > (M+1)^2$ makes the last expression greater than M. Thus the partial sums tend to $+\infty$ and the series diverges to $+\infty$. Individual terms tending to zero would not change this conclusion about the partial sums.

(c) The series with term $n!/n^n$.

Let $a_n = n!/n^n > 0$. The ratio simplifies as follows:

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} \\ &= \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n}.\end{aligned}$$

Exercise 12.14 justified $(1+1/n)^n \to e$ with finite $e \geq 2$. The reciprocal limit law therefore gives ratio limit $1/e$, which lies in $(0, 1)$. The Ratio Test proves absolute convergence.

Conclusion. Parts (a) and (c) converge absolutely. Part (b) diverges to $+\infty$.

14.5 Exercise 14.5: Linearity of series and failure of a termwise product formula

What we must show. If $\sum a_n = A$ and $\sum b_n = B$ are finite real sums, prove the two linearity formulas and examine the proposed product formula.

Preparation: use partial sums, whose limits are given.

Write $S_N = \sum_{n=1}^{N} a_n$ and $T_N = \sum_{n=1}^{N} b_n$. The hypotheses mean $S_N \to A$ and $T_N \to B$. These are finite limits, so the sum and constant multiple laws from Section 9 apply.

(a) Addition of two series.

Finite distributivity gives

$$\sum_{n=1}^{N} (a_n + b_n) = S_N + T_N.$$

Taking limits yields $A + B$. By the definition of a series sum, this proves $\sum_{n=1}^{\infty}(a_n + b_n) = A + B$.

(b) Multiplication by a fixed real constant.

For any $k \in \mathbb{R}$ and every N,

$$\sum_{n=1}^{N} ka_n = kS_N.$$

The right side tends to kA, including when $k = 0$ or $k < 0$. Hence $\sum_{n=1}^{\infty} ka_n = kA$.

(c) Test the termwise product formula on finite series.

Let $a_1 = 1$ and $a_n = 0$ for $n \geq 2$. Let $b_2 = 1$ and $b_n = 0$ for every $n \neq 2$. Each series has sum 1, so $AB = 1$. At $n = 1$ the second factor is zero; at $n = 2$ the first factor is zero; at every other index both are zero. Thus $a_n b_n = 0$ for all n, and

$$\sum_{n=1}^{\infty} a_n b_n = 0 \neq 1 = AB.$$

The conjectured identity is false even for series with only one nonzero term each. The finite product $S_N T_N$ contains all the terms $a_i b_j$ with $1 \leq i, j \leq N$, whereas the partial sum $\sum_{n=1}^{N} a_n b_n$ includes only the terms with equal indices. The product limit law applies to $S_N T_N$, not to that shorter sum.

Conclusion. The two linearity identities hold. The termwise product identity does not follow and is false in general.

14.6 Exercise 14.6: Absolutely summable terms times a bounded sequence

What we must show. Prove (a) that $\sum a_n b_n$ converges when $\sum |a_n|$ converges and (b_n) is bounded, and (b) recover Corollary 14.7.

(a), first step: put one bound on every second factor.

Boundedness supplies a finite $M > 0$ with $|b_n| \leq M$ for every n. We may insist on $M > 0$ by enlarging a given bound, even if every b_n is zero.

(a), second step: verify the Cauchy criterion for the product series.

Let $\varepsilon > 0$. Since the nonnegative series $\sum |a_n|$ converges, its Cauchy criterion gives an index N such that

$$n \geq m > N \quad \Longrightarrow \quad \sum_{j=m}^{n} |a_j| < \frac{\varepsilon}{M}.$$

For any such m, n, the finite triangle inequality and the bound on b_j imply

$$\left|\sum_{j=m}^{n} a_j b_j\right| \leq \sum_{j=m}^{n} |a_j|\,|b_j| \leq M \sum_{j=m}^{n} |a_j| < M\frac{\varepsilon}{M} = \varepsilon.$$

This is exactly the Cauchy criterion for $\sum a_n b_n$. Theorem 14.4 therefore proves convergence. The same estimate without the outer absolute value also proves the Cauchy criterion for $\sum |a_n b_n|$, so the product series actually converges absolutely.

(b) Use a constant second sequence.

Choose $b_n = 1$ for all n. This sequence is bounded and $a_n b_n = a_n$. Part (a) then says that convergence of $\sum |a_n|$ implies convergence of $\sum a_n$, precisely Corollary 14.7.

Conclusion. Every bounded multiplier preserves absolute summability; the constant multiplier 1 gives the cited corollary.

14.7 Exercise 14.7: Powers greater than one of a summable nonnegative sequence

What we must show. If $a_n \geq 0$, $\sum a_n$ converges, and $p > 1$, prove that $\sum a_n^p$ converges.

1. A convergent series has terms tending to zero.

Corollary 14.5 gives $a_n \to 0$. Hence there is N_0 such that $0 \leq a_n < 1$ for all $n > N_0$.

2. Compare the powers only where that comparison is justified.

If $a_n = 0$, then $a_n^p = 0 \leq a_n$. If $0 < a_n \leq 1$, the positive exponent $p-1$ gives $a_n^{p-1} \leq 1$. Multiplication by $a_n > 0$ yields

$$0 \leq a_n^p = a_n a_n^{p-1} \leq a_n \qquad (n > N_0).$$

The restriction to sufficiently late indices matters: a number larger than 1 has its size increased by taking a power $p > 1$.

3. Include the finite initial part.

Let $A = \sum_{n=1}^{\infty} a_n < +\infty$ and $C = \sum_{n=1}^{N_0} a_n^p$, both nonnegative and finite. For $N > N_0$,

$$\sum_{n=1}^{N} a_n^p = C + \sum_{n=N_0+1}^{N} a_n^p \leq C + \sum_{n=N_0+1}^{N} a_n \leq C + A.$$

For $N \leq N_0$, the partial sum is at most $C \leq C + A$ as well. The partial sums of $\sum a_n^p$ are increasing because their terms are nonnegative, and the displayed inequalities bound them above. The monotone sequence theorem proves a finite limit.

Conclusion. The series $\sum a_n^p$ converges for every real $p > 1$.

14.8 Exercise 14.8: A geometric-mean comparison for two nonnegative series

What we must show. If $a_n, b_n \geq 0$ and both $\sum a_n$ and $\sum b_n$ converge, prove convergence of $\sum \sqrt{a_n b_n}$.

1. Derive the needed pointwise inequality.

The square of any real number is nonnegative. Apply this to $\sqrt{a_n} - \sqrt{b_n}$, whose square expands to

$$0 \le (\sqrt{a_n} - \sqrt{b_n})^2 = a_n + b_n - 2\sqrt{a_n b_n}.$$

Rearranging and dividing by $2 > 0$ gives

$$0 \le \sqrt{a_n b_n} \le \frac{a_n + b_n}{2} \le a_n + b_n.$$

The last inequality uses $a_n + b_n \ge 0$. The square-root identity also holds when one or both of the numbers are zero.

2. Apply the Comparison Test to a convergent upper series.

By Exercise 14.5, $\sum(a_n + b_n)$ converges, with sum equal to the sum of the two original sums. Its terms are nonnegative. The displayed pointwise inequality therefore satisfies all the hypotheses of the Comparison Test, which proves convergence of $\sum \sqrt{a_n b_n}$.

Conclusion. The geometric-mean series converges. In fact its sum is at most one half of the sum of the two given series.

14.9 Exercise 14.9: Changing finitely many terms

What we must show. Suppose $a_n = b_n$ except for finitely many indices. Prove that $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

1. Place all exceptional indices before a single cutoff.

If the exceptional set is nonempty, let N_0 be its maximum. If the set is empty, take $N_0 = 1$. In either case $a_n = b_n$ for every $n > N_0$. Define the finite real constant

$$D = \sum_{n=1}^{N_0} (a_n - b_n).$$

2. Compare partial sums exactly.

For each $N \ge N_0$, all the later differences are zero, so

$$\sum_{n=1}^{N} a_n - \sum_{n=1}^{N} b_n = \sum_{n=1}^{N_0} (a_n - b_n) = D.$$

Thus the two partial-sum sequences differ by a fixed constant from that point onward.

3. Prove both directions of convergence.

If $\sum a_n = A$ is finite, the last identity gives $\sum_{n=1}^{N} b_n = \sum_{n=1}^{N} a_n - D \to A - D$, so the second series converges. Conversely, if $\sum b_n = B$ is finite, then $\sum_{n=1}^{N} a_n = \sum_{n=1}^{N} b_n + D \to B + D$. Therefore convergence of either series implies convergence of the other. If one diverges and the other converged, one of these implications would be contradicted; hence divergence is shared too.

Conclusion. Finite changes preserve convergence. They can change a finite sum by the constant D, which is why equality of sums is not asserted.

14.10 Exercise 14.10: Divergence detected by roots but not by ratios

What we must show. Give a series for which the Root Test proves divergence while neither conclusion of the Ratio Test applies.

1. Define positive terms with alternating adjacent ratios.

Take

$$a_n = 2^{\,n+(-1)^n} \qquad (n \geq 1).$$

Every term is positive, so its root and ratio are defined. The nth root is $a_n^{1/n} = 2 \cdot 2^{(-1)^n/n}$. Since $-1/n \leq (-1)^n/n \leq 1/n$ and powers of 2 preserve order,

$$2 \cdot 2^{-1/n} \leq a_n^{1/n} \leq 2 \cdot 2^{1/n}.$$

Both outer expressions tend to 2, using the root limit for the fixed positive number 2 and its reciprocal. The squeeze theorem gives $a_n^{1/n} \to 2$.

2. Apply the divergent part of the Root Test.

The root limit superior is $2 > 1$, so the Root Test proves divergence of $\sum a_n$.

3. Check both possible ratio-test inequalities.

When n is even, $(-1)^n = 1$ and $(-1)^{n+1} = -1$, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)-1-(n+1)} = 2^{-1} = \frac{1}{2}.$$

When n is odd, the two signs are reversed, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)+1-(n-1)} = 2^3 = 8.$$

Each value occurs infinitely often, and there are no other ratio values. Hence the ratio limit inferior is $1/2$ and its limit superior is 8. The convergence condition, ratio limit superior less than 1, fails. The divergence condition, ratio limit inferior greater than 1, fails as well. Thus the Ratio Test is inconclusive even though the Root Test settles the question.

Conclusion. The series $\sum 2^{n+(-1)^n}$ has the required properties: root limit 2, ratio limit inferior $1/2$, and ratio limit superior 8.

14.11 Exercise 14.11: A constant successive ratio gives a geometric series

What we must show. Suppose every $a_n \neq 0$ and $a_{n+1}/a_n = r$ is independent of n. Prove that $\sum a_n$ is geometric.

1. Convert the quotient statement to a recurrence.

Since $a_n \neq 0$, multiplication by a_n gives $a_{n+1} = ra_n$ for every n. Also $r = a_2/a_1 \neq 0$ because both a_1 and a_2 are nonzero.

2. Solve the recurrence by induction.

At $n = 1$, the formula $a_n = a_1 r^{n-1}$ reads $a_1 = a_1 r^0$, which holds. Suppose $a_n = a_1 r^{n-1}$ for some $n \geq 1$. Then

$$a_{n+1} = r a_n = r(a_1 r^{n-1}) = a_1 r^n.$$

This is the same formula with n replaced by $n + 1$, so induction proves it for every positive integer.

3. Recognize the series and its possible sum.

The series is therefore

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_1 r^{n-1} = a_1 + a_1 r + a_1 r^2 + \cdots,$$

which is the definition of a geometric series with first term a_1 and common ratio r. It converges when $|r| < 1$ and then has sum $a_1/(1 - r)$. If $|r| \geq 1$, the nonzero first term ensures that the terms do not tend to zero, so it diverges.

Conclusion. A constant successive ratio forces the exact geometric form $a_n = a_1 r^{n-1}$.

14.12 Exercise 14.12: Selecting an absolutely summable subsequence

What we must show. From $\liminf |a_n| = 0$, construct strictly increasing indices n_k for which $\sum_{k=1}^{\infty} a_{n_k}$ converges.

1. Arbitrarily small absolute values occur after every finite prefix.

Fix a number $\eta > 0$ and an index cutoff $N \geq 0$. If no $n > N$ satisfied $|a_n| < \eta$, every term beyond N would have $|a_n| \geq \eta$. All later tail infima would then be at least η, which would imply $\liminf |a_n| \geq \eta > 0$, a contradiction. Thus such an index always exists.

2. Select a geometric error bound at each stage.

Put $n_0 = 0$. After n_{k-1} has been chosen, use Step 1 with $N = n_{k-1}$ and $\eta = 2^{-k}$ to choose

$$n_k > n_{k-1}, \qquad |a_{n_k}| < 2^{-k}.$$

For definiteness, choose the least eligible index. Existence is provided by Step 1, and the least positive integer principle makes the recursive choice well defined. Strict increase of the indices ensures that (a_{n_k}) is a subsequence of the original sequence.

3. Compare the absolute-value series to a geometric series.

The geometric series satisfies $\sum_{k=1}^{\infty} 2^{-k} = 1$. Its terms dominate $|a_{n_k}| \geq 0$, so the Comparison Test gives convergence of $\sum |a_{n_k}|$. Absolute convergence implies convergence of $\sum a_{n_k}$ by Corollary 14.7. The geometric bound is essential to this argument: merely choosing a subsequence whose terms tend to zero would not by itself prove that its series converges.

Conclusion. There is an absolutely convergent subsequence series, obtained by imposing $|a_{n_k}| < 2^{-k}$ at stage k.

14.13 Exercise 14.13: Four exact sums from geometric and telescoping identities

What we must show. Calculate the two geometric sums in part (a), prove the identities in parts (b) and (c), and deduce the sum in part (d).

(a) The ratios $2/3$ and $-2/3$, starting at $n = 1$.

For any $|r| < 1$, shifting the index of the geometric series gives

$$\sum_{n=1}^{\infty} r^n = r \sum_{j=0}^{\infty} r^j = \frac{r}{1-r}.$$

Both ratios here have absolute value $2/3 < 1$. Substitution yields

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2/3}{1-2/3} = 2, \qquad \sum_{n=1}^{\infty} \left(-\frac{2}{3}\right)^n = \frac{-2/3}{1+2/3} = -\frac{2}{5}.$$

The negative sign in the second first term matters because its sum begins at exponent 1, rather than exponent 0.

(b) The series with term $1/[n(n+1)]$.

The finite algebraic identity

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)}$$

gives, for every $N \geq 1$,

$$\begin{aligned} \sum_{n=1}^{N} \frac{1}{n(n+1)} &= (1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \tfrac{1}{3}) + \cdots + (\tfrac{1}{N} - \tfrac{1}{N+1}) \\ &= 1 - \frac{1}{N+1}. \end{aligned}$$

Taking the limit $N \to \infty$ gives the finite sum 1.

(c) The series with term $(n-1)/2^{n+1}$.

Put the right side of the following identity over a common denominator to verify it:

$$\frac{n}{2^n} - \frac{n+1}{2^{n+1}} = \frac{2n-(n+1)}{2^{n+1}} = \frac{n-1}{2^{n+1}}.$$

The first term of the series, at $n = 1$, is zero. Summing the identity to N causes all intervening terms to cancel:

$$\sum_{n=1}^{N} \frac{n-1}{2^{n+1}} = \frac{1}{2} - \frac{N+1}{2^{N+1}}.$$

To justify the last term's limit explicitly, set $u_N = (N+1)/2^{N+1} > 0$. For $N \geq 1$,

$$\frac{u_{N+1}}{u_N} = \frac{N+2}{2(N+1)} \leq \frac{3}{4},$$

because the last inequality is equivalent to $N \geq 1$. Induction gives $0 < u_N \leq u_1(3/4)^{N-1}$, whose upper bound tends to zero. Hence $u_N \to 0$ by the squeeze theorem, and

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(d) Deduce the sum with term $n/2^n$.

For each index,

$$\frac{n}{2^n} = 2\frac{n-1}{2^{n+1}} + \frac{1}{2^n}.$$

The first series on the right converges by part (c), and the second is geometric with sum 1. The linearity formulas of Exercise 14.5 therefore apply and give

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2\left(\frac{1}{2}\right) + 1 = 2.$$

Conclusion. The requested sums are, in order, 2, $-2/5$, 1, $1/2$, and 2. All telescoping cancellations were made in finite partial sums before taking a limit.

14.14 Exercise 14.14: A blockwise lower comparison for the harmonic series

What we must show. Prove divergence of $\sum_{n=1}^{\infty} 1/n$ by comparing it with the displayed repeated-block sequence.

1. Express the displayed sequence by exact index ranges.

For each integer $j \geq 1$, the block

$$2^{j-1} \leq n \leq 2^j - 1$$

contains $2^j - 1 - 2^{j-1} + 1 = 2^{j-1}$ indices. On that block define $a_n = 2^{-j}$. Thus the first block is the single term $1/2$; the second contains two terms $1/4$; the third contains four terms $1/8$; and the fourth contains eight terms $1/16$. These blocks exactly reproduce the sequence in the question.

2. Every complete block contributes the same positive amount.

For the block labeled j, its sum is

$$2^{j-1} \cdot 2^{-j} = \frac{1}{2}.$$

The comparison series in the exercise begins at $n = 2$, so it omits the first block and retains all blocks with $j \geq 2$. At the end of block $J \geq 2$ its partial sum is

$$\sum_{n=2}^{2^J-1} a_n = \sum_{j=2}^{J} \frac{1}{2} = \frac{J-1}{2}.$$

These values tend to $+\infty$ as J grows. All intermediate partial sums are increasing, since $a_n > 0$. Therefore the entire comparison series, not only these block-end partial sums, diverges to $+\infty$.

3. Compare each harmonic term with its block value.

Given $n \geq 2$, there is a unique block index $j \geq 2$ with $2^{j-1} \leq n \leq 2^j - 1$. Since $n < 2^j$ and both are positive,

$$\frac{1}{n} > \frac{1}{2^j} = a_n.$$

The Comparison Test now shows that $\sum_{n=2}^{\infty} 1/n$ diverges to $+\infty$. Adding the first harmonic term, which is 1, preserves this divergence.

Conclusion. The harmonic series diverges to $+\infty$ because it dominates a positive series whose every complete block adds $1/2$.